

THEORY OF COMPUTATION (CS302)

Complete Syllabus

Course Description

Theory of Computation provides the mathematical foundation of computer science. It focuses on automata, formal languages, grammars, computability, and Turing machines.

Course Objectives

- Understand automata theory fundamentals.
- Study DFA and NFA models.
- Learn regular languages and regular expressions.
- Understand CFGs, PDAs, and Turing Machines.
- Analyze decidability and computability.

Unit 1 - Finite Automata

DFA, NFA, Transition Diagrams, Transition Tables, Language Recognition and Applications.

Unit 2 - Regular Languages

Regular Expressions, Regular Languages, Pumping Lemma and Applications.

Unit 3 - Context-Free Grammars

CFGs, Derivations, Parse Trees, Ambiguity, CNF and Grammar Simplification.

Unit 4 - Pushdown Automata

PDA Components, Stack Operations, Acceptance Methods and PDA-CFG Equivalence.

Unit 5 - Turing Machines & Decidability

Turing Machines, Recursive Languages, Recursively Enumerable Languages, Halting Problem and Decidability.

Course Outcomes

Students will be able to design automata, construct grammars, analyze formal languages and understand computability concepts.

Recommended Textbooks

Hopcroft, Motwani & Ullman; Peter Linz.

Summary

This course establishes the theoretical foundations required for advanced computer science subjects.